Computer Science & IT



Fundamental of C Language

Data Types & Operator

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



Topic — System program

Topic — preprocessor . # define Size 100

Topic — Complier, Assembler,

Topic — Linker & Loader

Topic — object file / Exe file

Slide

Topics to be Covered



Topic

Topic

Topic

Topic

Topic

Number system

Binary No Conversion

Singed No. & unsigned No

Slide



Macro

Macro Small piece of code which can be reused

#define size 100

Wherever in program Size is written it replace it by 100



Macro

- Macros in C are powerful tools that allow developers to define reusable code snippets.



Macro

- They are defined using preprocessor directives and primarily used for code substitution and generation.

#define
#include



Macro

- Macros provide a convenient way to write concise and efficient code, improving the readability and maintainability of C programs.



Topic : Number System

Decimal No. System : $425 \longrightarrow 4\times10^2 + 2\times10^1 + 5\times\underline{1}0^0$

$400+20+5$

(0, 1, 2, 3, 4, 5, 6, 7, 8, 9)

Binay No. System

$\underline{1}0\underline{1}00$

(0, 1)

Slide



Topic : Binary Number

10100

Binay No ⟶ Decimal No. Convorsion

1 0 1 0 0 = $1\times2^4+0\times2^3+1\times2^2+0\times2^1+0\times2^0$

$= 16+0+4+0+0$

$= 20$

2^4 2^3 2^2 2^1 2^0

Slide

10100 Binay No Decimal value 20

Convert Binay to decimal.

(A) $\underset{3\ 2\ 1\ 0}{1\ 1\ 1\ 1} = 1\times2^3 + 1\times2^2 + 1\times2^1 + 1\times2^0 = 8+4+2+1 = \underline{15}$

(B) $\underset{4\ 3\ 2\ 1\ 0}{1\ 1\ 0\ 0\ 1} = 1\times2^4 + 1\times2^3 + 0\times2^2 + 0\times2^1 + 1\times2^0$

$16+8+0+0+1 = 25$



Topic : Binary Number Interpretation

(A) $\overbrace{1111\ldots1}^{n\text{-times}}$ =

(B) 12012

Teonary No then what is the decimal equivalent



Topic : Binary Number Interpretation

$2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

$2^5 = 32$

$2^6 = 64$

$2^7 = 128$

$2^8 = 256$

$2^9 = 512$

$2^{10} = 1024$

$2^{11} = 2048$

$2^{12} = 4096$

$2^{13} = 8192$

$2^{14} = 16384$

$2^{15} = 32768$

$2^{16} = 65536$

Slide



Topic : Binary Number Conversion

for a given Decimal No ⟶ Binay No.

$\frac{7}{3}$ Quotient = 2

Remainder = 1

Question: Convert Decimal 7 to binay

2	7	
2	3	1
2	1	1
	0	1

↑

1 1 1

Binay No. for 7

Slide



Topic : Binary Number Conversion

$$\underset{2}{1}\ \underset{1}{1}\ \underset{0}{1} \longrightarrow 1\times2^2+1\times2^1+1\times2^0 = 4+2+1 = 7$$

Binary $\longrightarrow$ Decimal

$$\underset{\text{MSB}}{1}\,0\,1\,1\,\underset{\text{LSB}}{0}$$

Binay No.

Most significant bit

Least significant bit

Slide



Topic : Binary Number Conversion

Q. convert Decimal 29 to binay

2 | 29		
2 | 14	1	
2 | 7	0	
2 | 3	1	
2 | 1	1	
0	1	



1 1 1 0 1

4 3 2 1 0

$1\times2^4+1\times2^3+1\times2^2+0\times2^1+1\times2^0$

$16+8+4+0+1=29$

Slide



Topic : Binary Number Conversion

Q. Convert 127 to Binary No.

Ans

2	127	
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

1 1 1 1 1 1 1

Slide



Topic : Binary Number Conversion

Convert the following Decimal to Binary

(A) 428

(B) 313

(C) 194

(D) 98

1 1 0 0 0 1 0

64 + 32 + 0 + 0 + 0 + 2 + 0

Slide



Topic : Unsigned & signed Number

integer – positive No & Negative

C language integer is of 2 types

- Unsigned — only positive No.
- Singed No — positive No. & Negative No.

Slide



Topic : Unsigned Integer Representation

Unsinged No representation

10 ⟶ Binay Conversion ⟶ 1010

2	10	
2	5	0
2	2	1
2	1	0
	0	1

Unsigned representation

Slide



Topic : Signed Integer Representation

positive No

Negative No

How to differenciate between positive No. & Negative No.

Ans:- For representing the Sign 1 bit (MSB) is reserved.

if MSB is 0 then No is positive

if MSB is 1 then No is Negative

Slide



Topic : Signed Interger Representation

2's complement representation

4 bits Decimal - 2 : 0 0 1 0

$$
\begin{array}{r|l}
2 & 2 \\
\hline
2 & 1 \quad 0 \\
\hline
 & 0 \quad 1
\end{array}
$$

what is -2 :- Bit wise complement

0 0 1 0

↓ ↓ ↓ ↓

1 1 0 1

(complement

0 → 1

1 → 0)

Slide



Topic : 2's complement representation

(1) $+2 \; - \; 0010$

(2) Bit wise complement $\; 0010$

$\downarrow\downarrow\downarrow\downarrow$

1101

$$\begin{array}{r} 0 \\ +\,0 \\ \hline 0 \end{array} \qquad \begin{array}{r} 0 \\ +\,1 \\ \hline 1 \end{array} \qquad \begin{array}{r} 1 \\ +\,0 \\ \hline 1 \end{array}$$

(3) Addition of 1 $\; 1101$

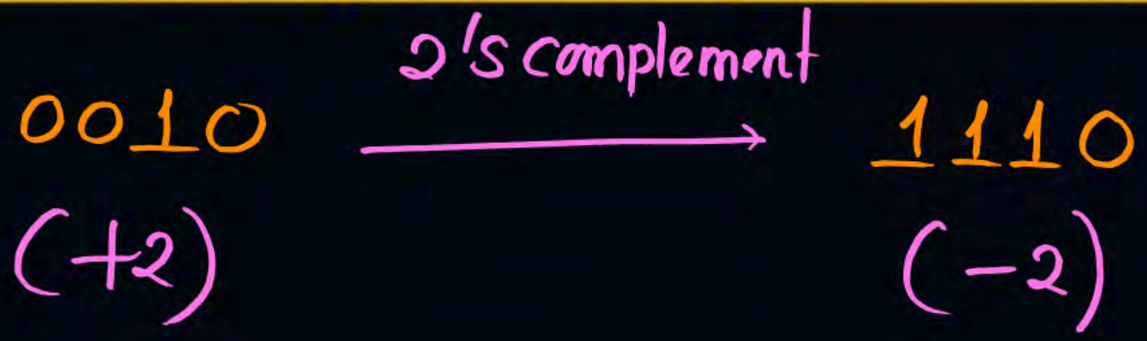
$$+ \; \begin{array}{r} 1 \\ 1 \\ \hline 1110 \end{array}$$

$$\begin{array}{r} 1 \\ 1 \\ \hline 10 \\ \uparrow \end{array}$$

Slide



Topic : 2's complement representation



Slide



Topic : 2's complement representation

$+4\ (0100) \xrightarrow{\text{2's complement}}$

1011 (6ht wise complement)

$+\quad 1\ 1^{1}$ (Addition of 1)

1100 (-4)

Slide



Topic : 2's complement representation

$+7\ (0111) \xrightarrow{2\text{'s complement}}$

1000 (bitwise complement)

$\underline{\quad\quad 1}$ (Addition of 1's)

$1001\ (-7)$

Slide



Topic : 2's complement representation

The given No 1011 is a Signed No 2 in 2's complement form, what is the decimal value?

(-5) 1011 ⟶ MSB is 1 then its a Negative No.

take 2's complement – 1011 ⟶ 0100 (bit wise complement)

$$\begin{array}{r} 0100 \\ +\quad\ 1 \\ \hline 0101\ (+5) \end{array}$$ (Add 1)

$0\times2^3+1\times2^2+0\times2^1+1\times2^0=5$

Slide



Topic : 2's complement representation

1011 Negative No., 2's complement form

↓

Expansion of 2's complement Negative No.

1011

3210

MSB weight is Negative

$-1\times2^3+0\times2^2+1\times2^1+1\times2^0$

$-8+0+2+1 = -8+3 = \boxed{-5}$

Slide



Topic : 2's complement representation

Question: Suppose No 11101 is Sing No in 2's complement No. what is the decimal value of it.

Explanation:

$$\underset{4}{1}\ \underset{3}{1}\ \underset{2}{1}\ \underset{1}{0}\ \underset{0}{1}$$

MSB is 1 Hence No. is Negative

$$-1\times2^4+1\times2^3+1\times2^2+0\times2^1+1\times2^0$$

$$-16+8+4+0+1 = -16+13 = (-3)$$

Slide



Topic : 2's complement representation

Size in Computer is 1 bit 2 Byte

0 — 1 bit

1 ⟶ 1 bit

example : 1100 4 bits

Group of 8 bits is called Byte



← 8 bits or 1 Byte

Slide



Topic : 2's complement representation

int Size is 4Bytes , $4 \times 8 = 32$ bits

Slide



Topic : Octal Number

Octal No. are those No. having base 8

(0,1,2,3,4,5,6,7)

$(57)_8$

octal

$(47)_{10} \rightarrow$

8	47	
8	5	7
	0	5

5 7

1 0

$5 \times 8^1 + 7 \times 8^0$

$40 + 7 = 47$

Slide



Topic : Octal Number

Octal No. are those No. having base 8

$(117)_{10}$

Convert to octal

8	117	
8	14	5
8	1	6
	0	1

$(165)_8$

$1\times8^2+6\times8^1+5\times8^0$

$64+48+5 = 117$

Slide



Topic : Hexadecimal Number

Hexa Decimal Base 16

(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)

a, b, c, d, e, f

$(64)_{10}$

16 | 64

16 | 4 — 0

0 — 4

$(40)_{16} = 4 \times 16^1 + 0 \times 16^0$

$= 64 + 0 = 64$

Slide



Topic : Hexadecimal Number

$(92)_{10} \longrightarrow$ Hexadecimal

16 | 92		
16 | 5	C	↑
0	5	↑

$(5C)_{16}$

$5 \times 16^1 + 12 \times 16^0 = 80 + 12 = \underline{92}$

Slide



2 mins Summary

Topic — Binay No.

Topic — Unsigned Conversion expansion

Topic — Singed No. 2's complement

Topic — octal No. Hexadecimo No.

Topic

Slide

THANK - YOU

Slide